Coding Challenge

Domain: College Student Management System

Problem Statement

Assume that you are an Data Analyst you need to analyze a college which maintains students, courses, marks.

The table structure is as follows:

```
-- Students Table
CREATE TABLE students (
   student_id INT PRIMARY KEY,
   student_name VARCHAR(50),
   gender VARCHAR(10),
   city VARCHAR(50),
   join_year INT
);

-- Courses Table
CREATE TABLE courses (
   course_id INT PRIMARY KEY,
   course_name VARCHAR(100),
   department VARCHAR(50)
);

-- Marks Table
CREATE TABLE marks (
   mark_id INT PRIMARY KEY,
   student_id INT,
   course_id INT,
   marks INT,
   FOREIGN KEY (student_id) REFERENCES students(student_id),
   FOREIGN KEY (course_id) REFERENCES courses(course_id)
);
```

- **Step 2: Insert Data**

```
-- Insert into students
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES
(1, 'Anu', 'F', 'Tumakuru', 2024),
(2, 'Ravi', 'M', 'Bengaluru', 2023),
(3, 'Kiran', 'M', 'Tumakuru', 2024),
(4, 'Sneha', 'F', 'Mysuru', 2023),
(5, 'Manu', 'M', 'Tumakuru', 2022);

-- Insert into courses
INSERT INTO courses (course_id, course_name, department) VALUES
(101, 'SQL Basics', 'Computer Science'),
(102, 'Excel for Analysts', 'Commerce'),
(103, 'Statistics', 'Mathematics');

-- Insert into marks
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES
(1, 1, 101, 85),
(2, 2, 101, 72),
(3, 3, 101, 90),
(4, 4, 102, 88),
(5, 5, 103, 67),
(6, 1, 103, 79),
(7, 2, 102, 81);
```

Level 1: Very Basic

1. Display all students.
2. Display only student_name and city from students table.
3. Show all courses.
4. Display students who are from Tumakuru.
5. Display students who joined in 2024.

Level 2: Filtering

6. Show students whose gender is F.
7. Show marks greater than 80.
8. Display course names from Commerce department.

9. Show students who are **not** from Bengaluru.
10. Display marks between 70 and 90.

Level 3: Sorting

11. Display all students ordered by student_name ascending.
12. Show marks ordered from highest to lowest.
13. Display students ordered by join_year descending.

Level 4: Aggregate Basics

14. Find total number of students.
15. Find average marks.
16. Find highest marks.
17. Find lowest marks.
18. Find total marks scored by all students.